Class Activity-1 (20/07/2026)

1.Hospital Patient Admission Analysis

```
library(ggplot2)

library(treemap)

# Hospital Patient Admission Dataset

hospital <- data.frame(

  Department = c("Cardiology","Neurology","Pediatrics","Orthopedics","Emergency"),

  Admissions = c(150,120,180,100,250),

  Month = c("Jan","Jan","Jan","Jan","Jan"))

# 1. Bar Chart

ggplot(hospital, aes(x = Department, y = Admissions, fill = Department)) +

  geom_bar(stat = "identity") +

  labs(title = "Hospital Patient Admissions",

       x = "Department",

       y = "Admissions")

# 2. Stacked Bar Chart

ggplot(hospital, aes(x = Month, y = Admissions, fill = Department)) +

  geom_bar(stat = "identity") +

  labs(title = "Stacked Bar Chart",

       x = "Month",

       y = "Admissions")

# 3. Treemap

treemap(

  hospital,

  index = "Department",

  vSize = "Admissions",

  title = "Hospital Admissions Treemap"

)
```

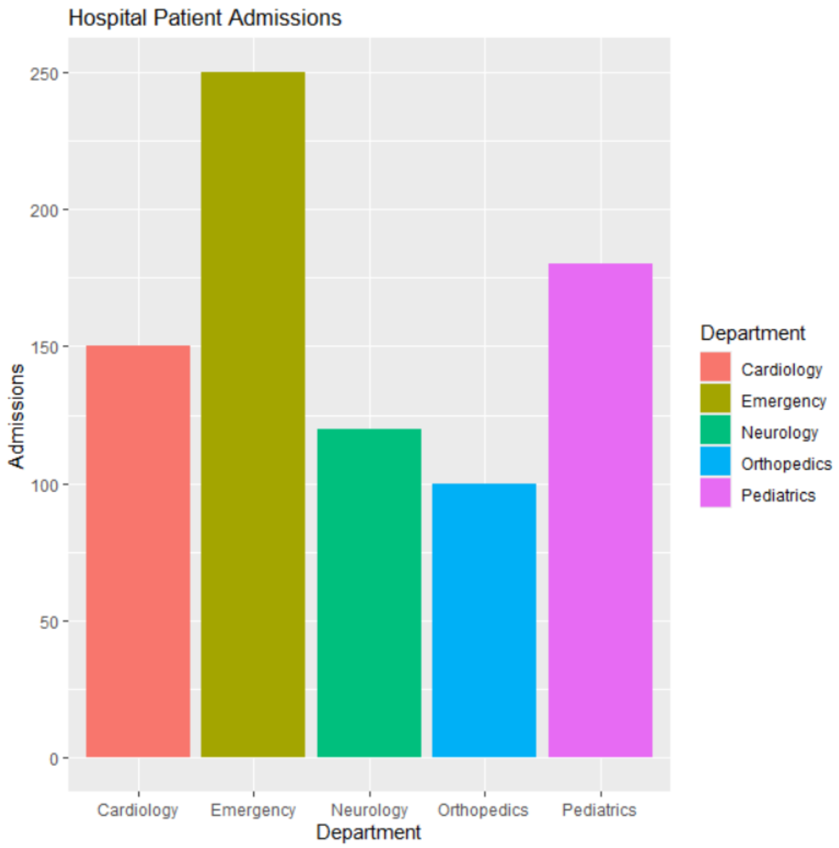
4. Box Plot

```
ggplot(hospital, aes(x = Department, y = Admissions, fill = Department)) +
  geom_boxplot() +
  labs(title = "Admission Variability",
       x = "Department",
       y = "Admissions")
```

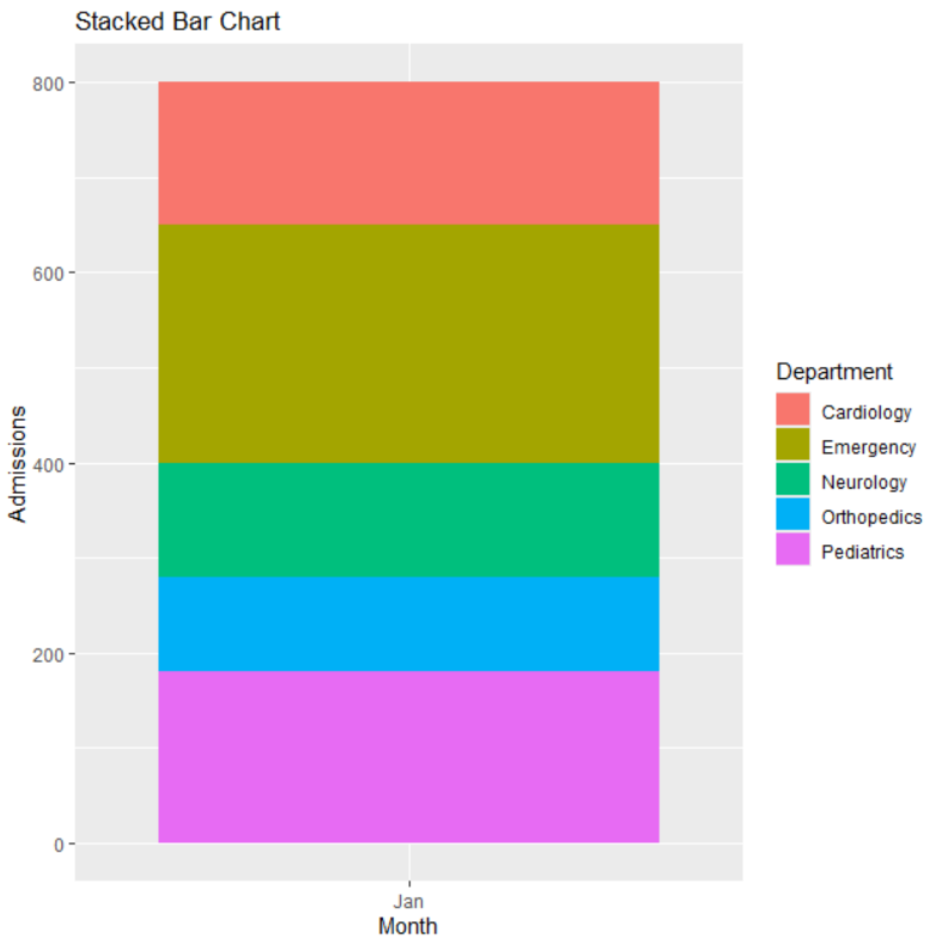
5. Histogram

```
ggplot(hospital, aes(x = Admissions)) +
  geom_histogram(binwidth = 20, fill = "skyblue", color = "black") +
  labs(title = "Distribution of Admissions",
       x = "Admissions",
       y = "Frequency")
```

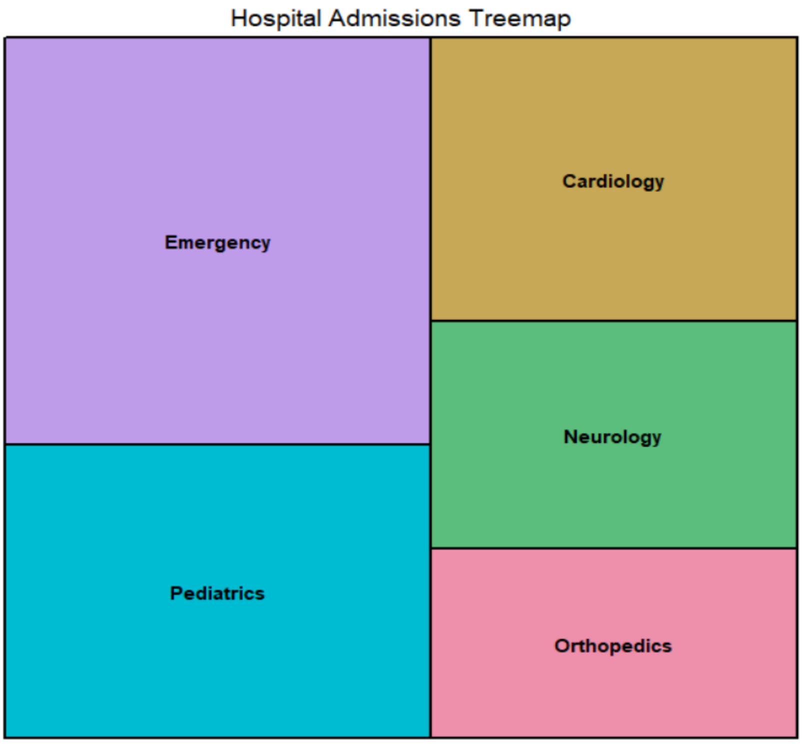
1. Bar Chart



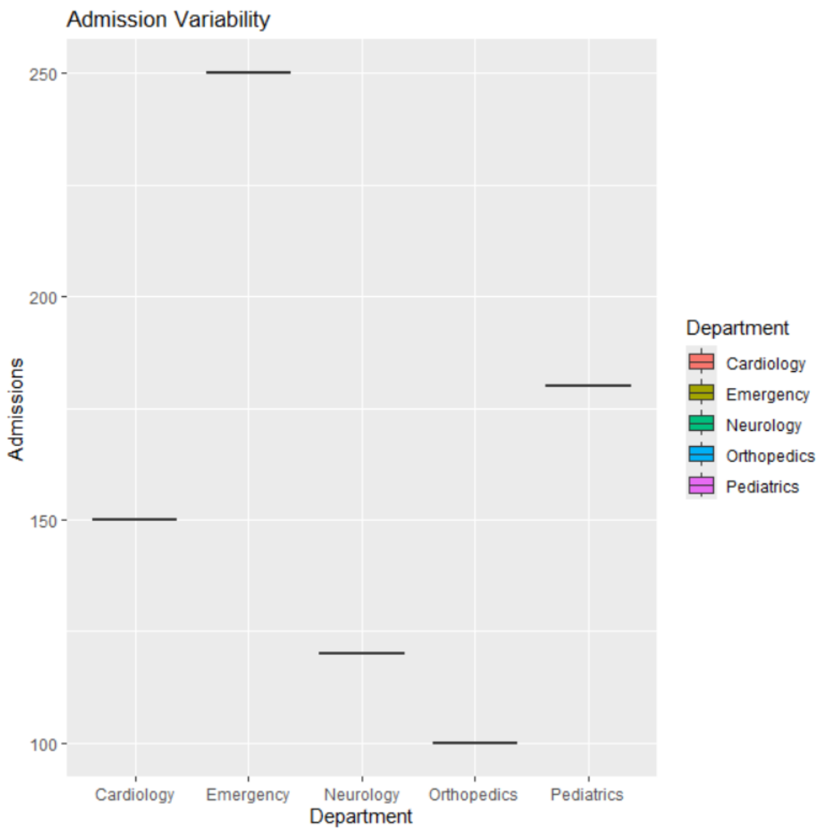
2. Stacked Bar Chart



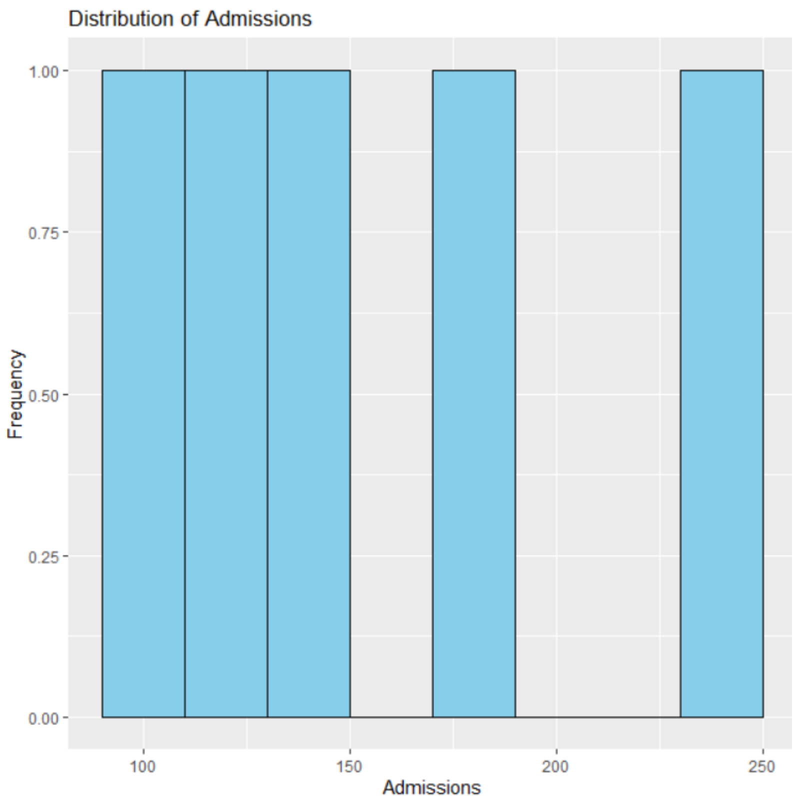
3. Treemap



4. Box Plot



5. Histogram



2.E-Commerce Customer Spending Behaviour.

```r
library(ggplot2)
# E-Commerce Customer Spending Data
customer <- data.frame(
  Customer = c("C1","C2","C3","C4","C5","C6","C7","C8","C9","C10"),
  Spending = c(500,700,1200,900,1500,800,2000,1300,600,1000),
  Orders = c(2,3,5,4,6,3,7,5,2,4),
  Category = c("Electronics","Clothing","Electronics","Groceries",
               "Electronics","Clothing","Groceries","Electronics",
               "Clothing","Groceries")
)

# 1. Histogram
ggplot(customer, aes(x = Spending)) +
  geom_histogram(binwidth = 300, fill = "skyblue", color = "black") +
  labs(title = "Histogram of Customer Spending",
       x = "Spending",
       y = "Frequency")

# 2. Violin Plot
ggplot(customer, aes(x = Category, y = Spending, fill = Category)) +
  geom_violin() +
  labs(title = "Violin Plot of Spending by Category",
       x = "Category",
       y = "Spending")

# 3. Box Plot
ggplot(customer, aes(x = Category, y = Spending, fill = Category)) +
  geom_boxplot() +
```

```
  labs(title = "Box Plot of Spending by Category",
       x = "Category",
       y = "Spending")
```

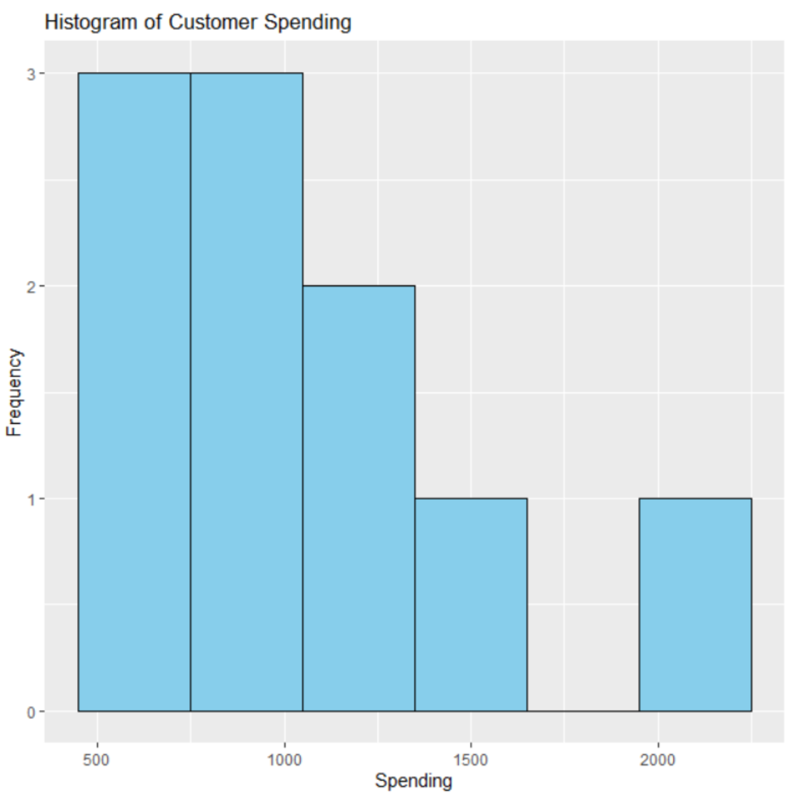
4. Density Plot

```
ggplot(customer, aes(x = Spending)) +
  geom_density(fill = "lightgreen", alpha = 0.5) +
  labs(title = "Density Plot of Customer Spending",
       x = "Spending",
       y = "Density")
```

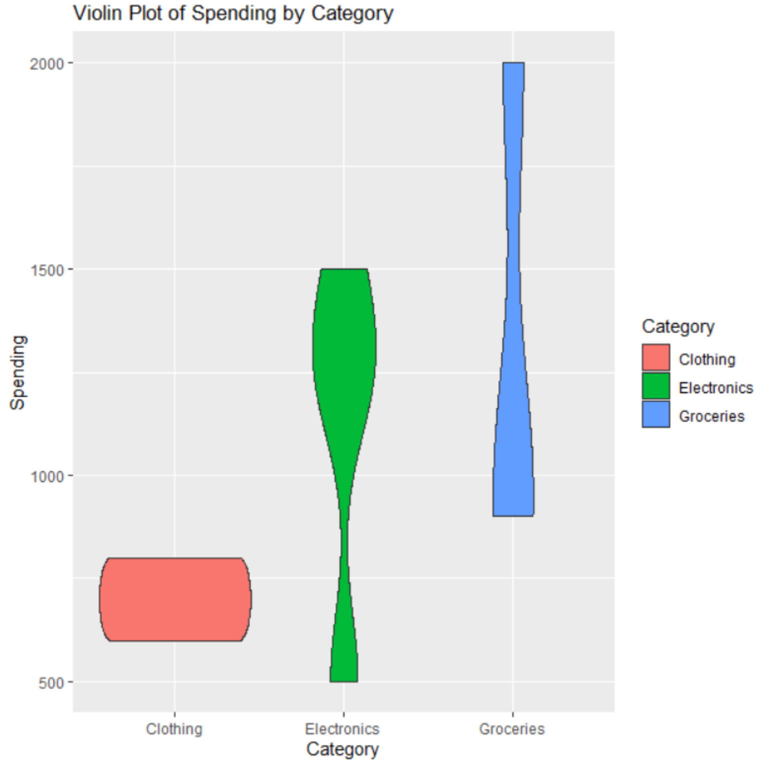
5. Bar Chart

```
ggplot(customer, aes(x = Category, fill = Category)) +
  geom_bar() +
  labs(title = "Customer Count by Product Category",
       x = "Category",
       y = "Number of Customers")
```

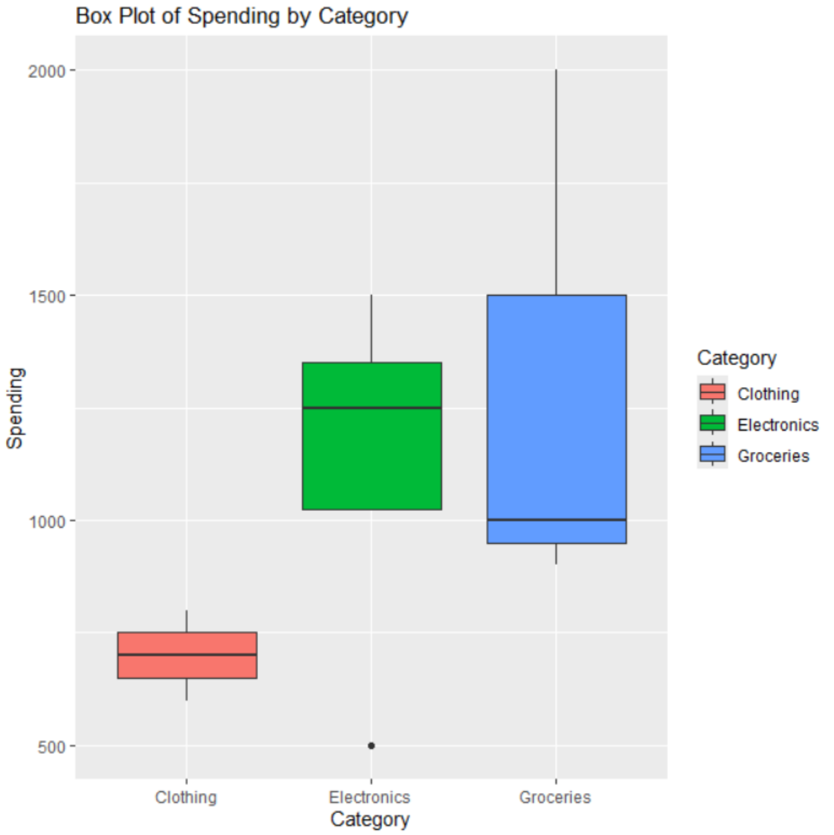
1. Histogram



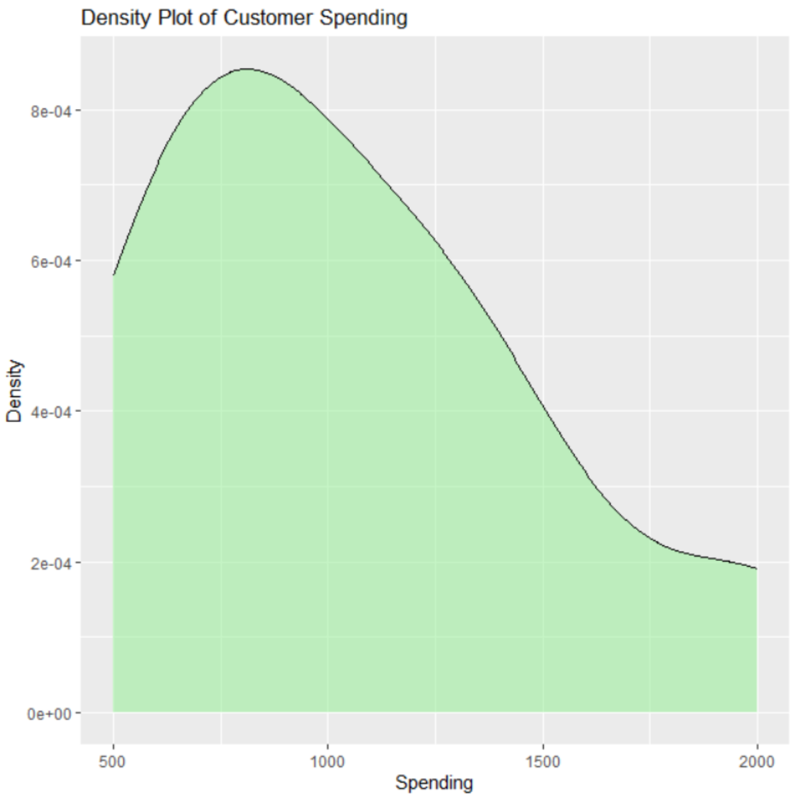
2. Violin Plot



3. Box Plot



4. Density Plot



5. Bar Chart

3.University Examination Performance Dashboard

```
library(ggplot2)
# University Examination Dataset
student <- data.frame(
  Department = c("CSE","ECE","MECH","CIVIL","AI&DS",
                 "CSE","ECE","MECH","CIVIL","AI&DS"),
  Class = c("A","A","A","A","A","B","B","B","B","B"),
  Marks = c(85,78,65,72,90,88,80,70,75,95))
```

 # 1. Bar Chart - Average Marks by Department

```
ggplot(student, aes(x = Department, y = Marks, fill = Department)) +
  geom_bar(stat = "identity") +
  labs(title = "Marks by Department",
       x = "Department",
       y = "Marks")
```

 # 2. Box Plot - Score Distribution and Outliers

```
ggplot(student, aes(x = Department, y = Marks, fill = Department)) +
  geom_boxplot() +
  labs(title = "Marks Distribution by Department",
       x = "Department",
       y = "Marks")
```

 # 3. Histogram - Marks Distribution

```
ggplot(student, aes(x = Marks)) +
  geom_histogram(binwidth = 5, fill = "skyblue", color = "black") +
  labs(title = "Distribution of Student Marks",
       x = "Marks",
       y = "Frequency")
```

 # 4. Violin Plot - Marks Distribution

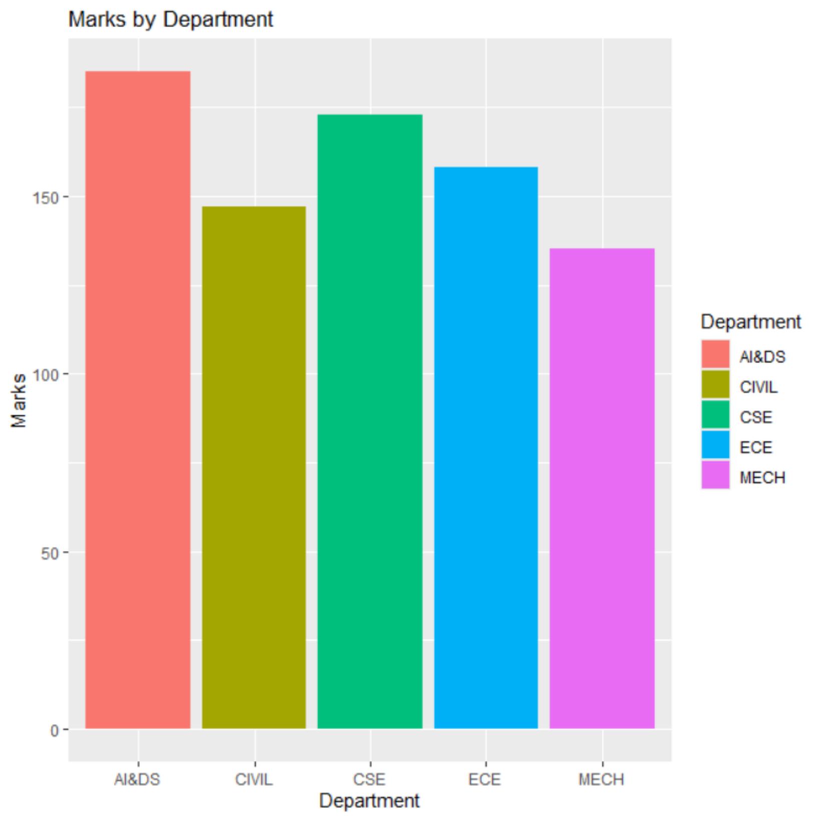
```
ggplot(student, aes(x = Department, y = Marks, fill = Department)) +
```

```
  geom_violin() +
  labs(title = "Marks Distribution (Violin Plot)",
       x = "Department",
       y = "Marks")
```

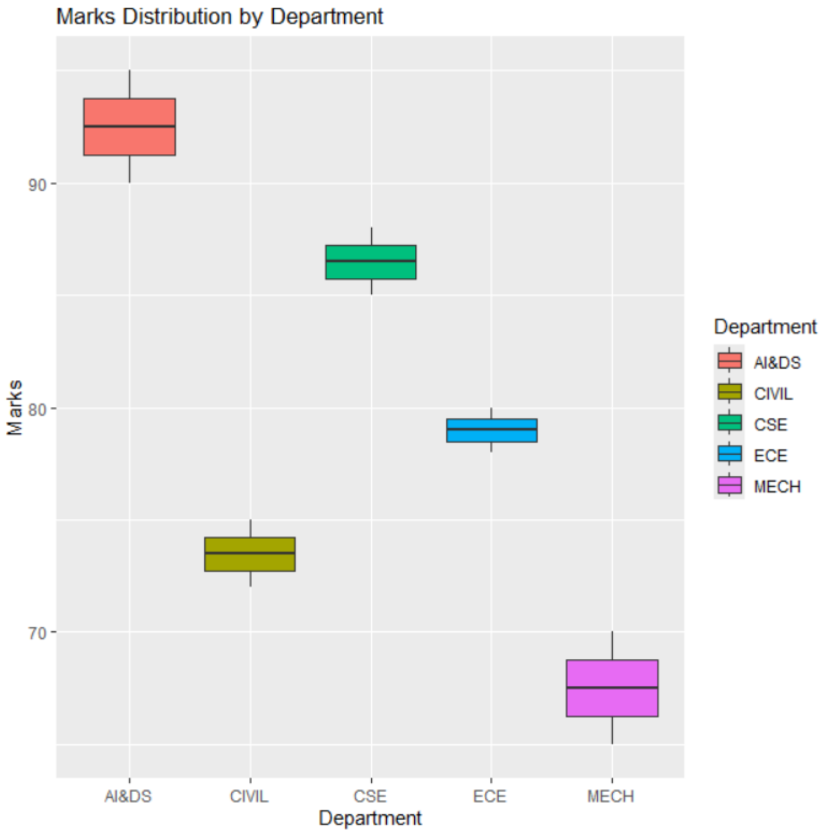
5. Scatter Plot - Top Performing Classes

```
ggplot(student, aes(x = Class, y = Marks, color = Department)) +
  geom_point(size = 4) +
  labs(title = "Top Performing Classes",
       x = "Class",
       y = "Marks")
```

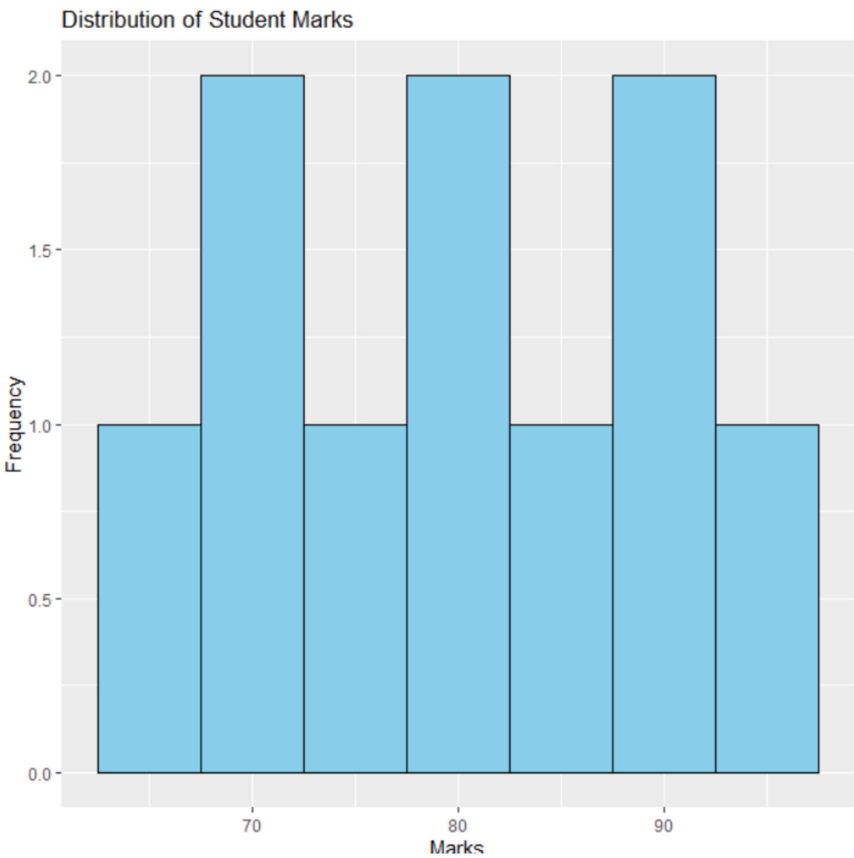
1. Bar Chart - Average Marks by Department



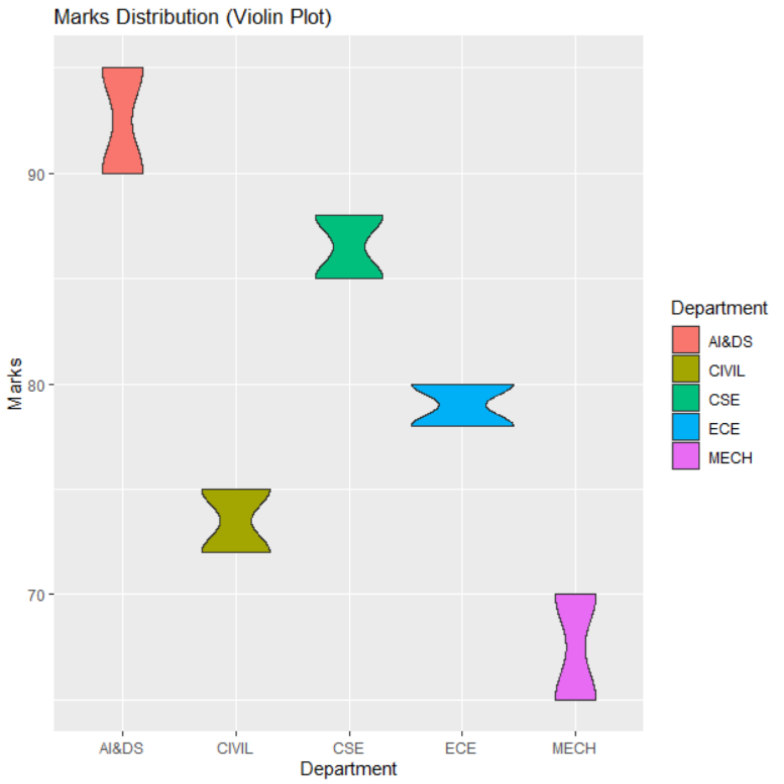
2. Box Plot - Score Distribution and Outliers



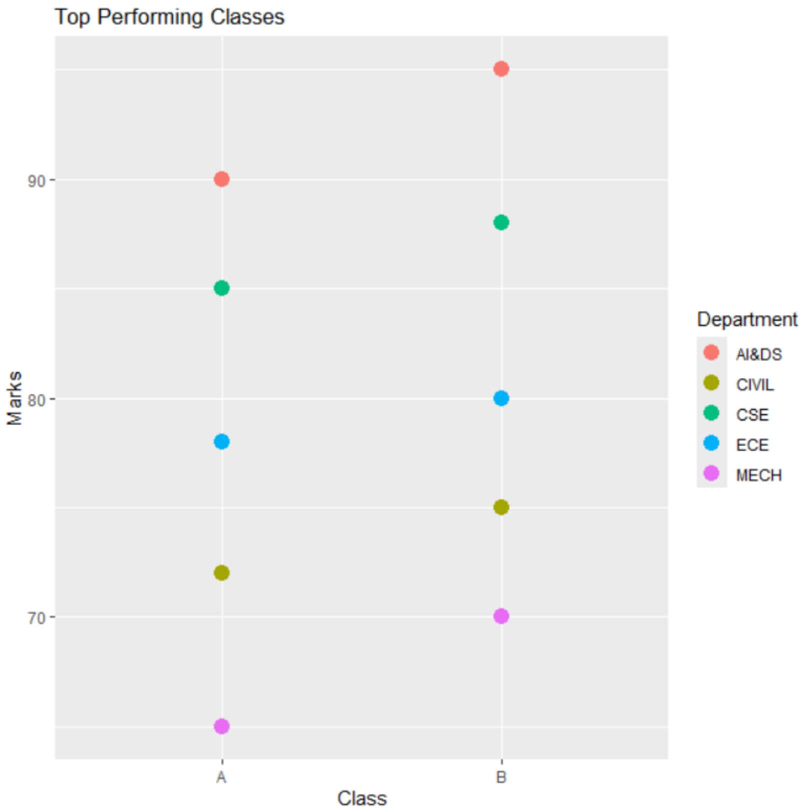
3. Histogram - Marks Distribution



4. Violin Plot - Marks Distribution



5. Scatter Plot - Top Performing Classes

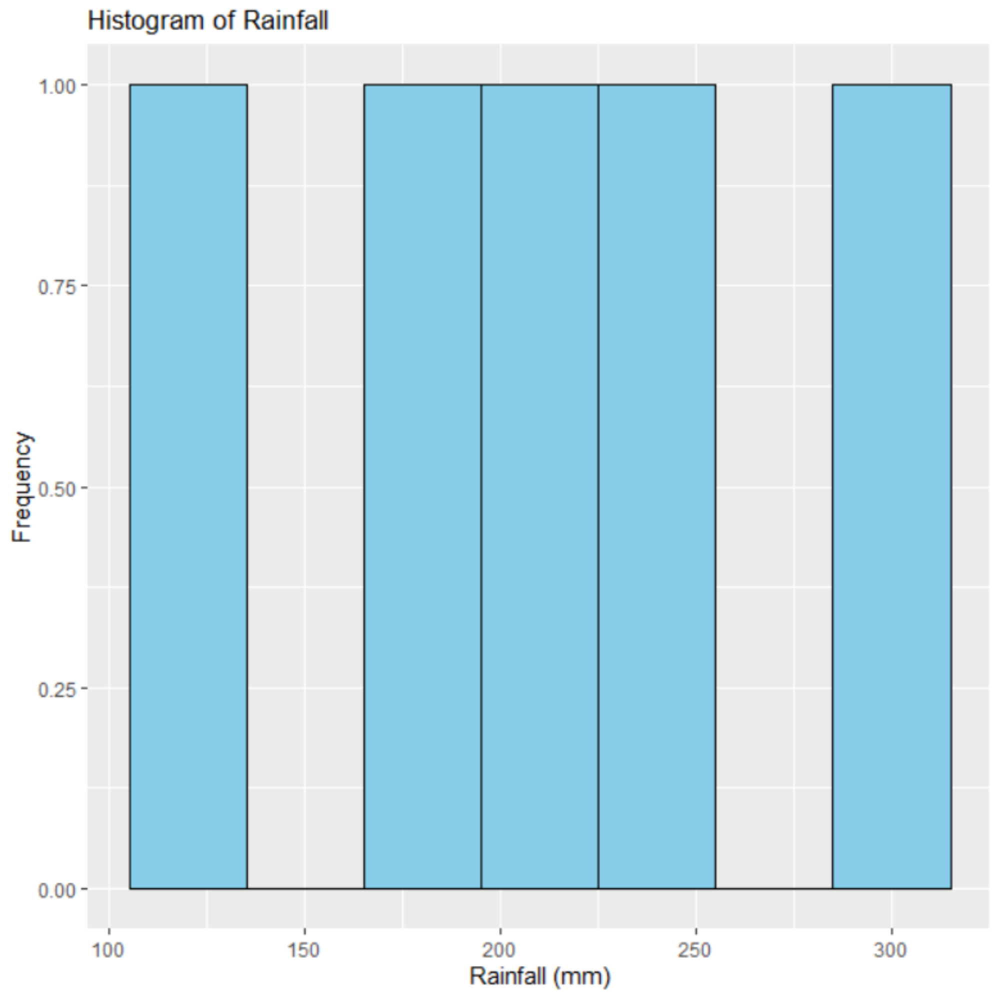


4. Climate and Rainfall Distribution Study.

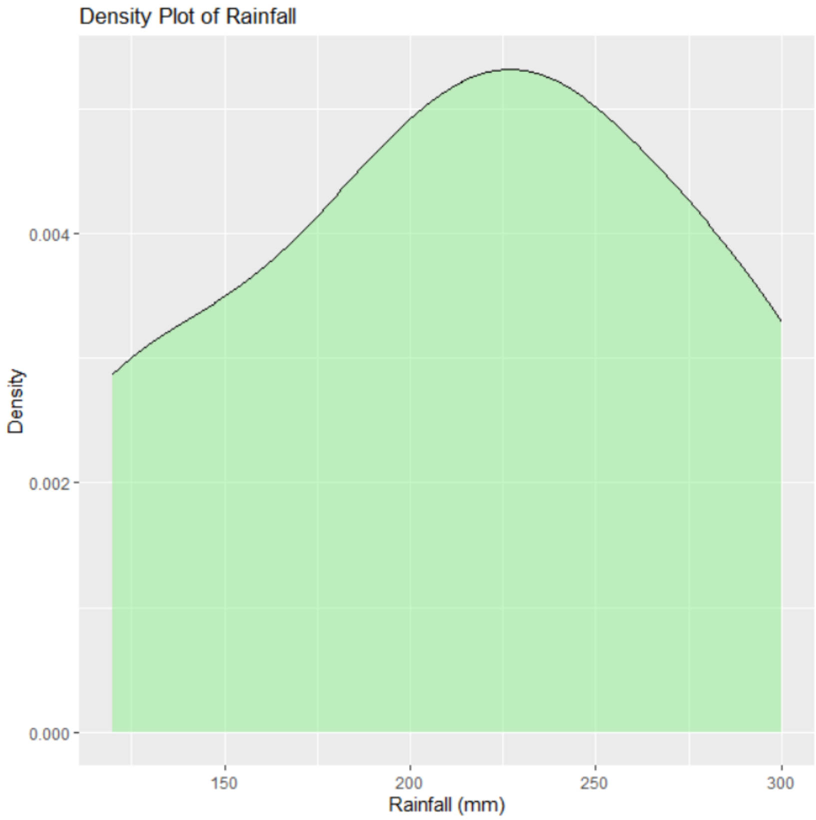
```r
library(ggplot2)
# Climate Dataset
climate <- data.frame(
  City = c("Chennai","Bangalore","Mumbai","Delhi","Kolkata"),
  Rainfall = c(220,180,300,120,250),
  Temperature = c(34,28,31,36,33),
  Humidity = c(80,70,85,60,78)
)
# 1. Histogram - Rainfall Distribution
ggplot(climate, aes(x = Rainfall)) +
  geom_histogram(binwidth = 30, fill = "skyblue", color = "black") +
  labs(title = "Histogram of Rainfall",
       x = "Rainfall (mm)",
       y = "Frequency")
# 2. Density Plot - Rainfall Distribution
ggplot(climate, aes(x = Rainfall)) +
  geom_density(fill = "lightgreen", alpha = 0.5) +
  labs(title = "Density Plot of Rainfall",
       x = "Rainfall (mm)",
       y = "Density")
# 3. Box Plot - Rainfall by City
ggplot(climate, aes(x = City, y = Rainfall, fill = City)) +
  geom_boxplot() +
  labs(title = "Box Plot of Rainfall",
       x = "City",
       y = "Rainfall (mm)")
# 4. Bar Chart - Rainfall by City
ggplot(climate, aes(x = City, y = Rainfall, fill = City)) +
```

```
  geom_bar(stat = "identity") +
  labs(title = "Monthly Rainfall by City",
       x = "City",
       y = "Rainfall (mm)")
```

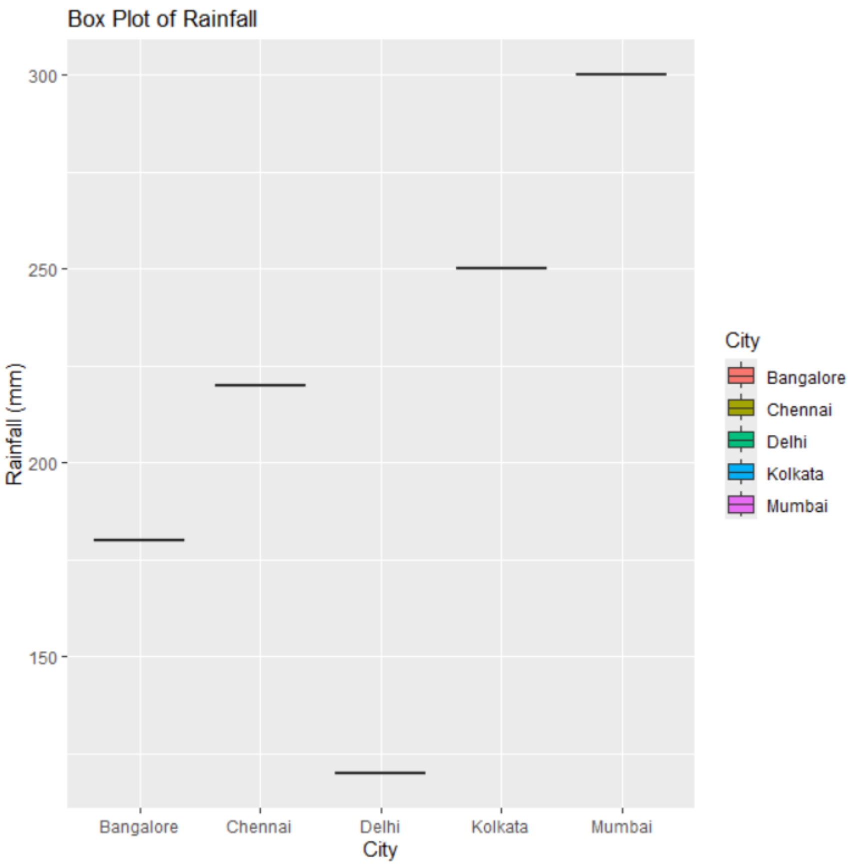
1. Histogram - Rainfall Distribution



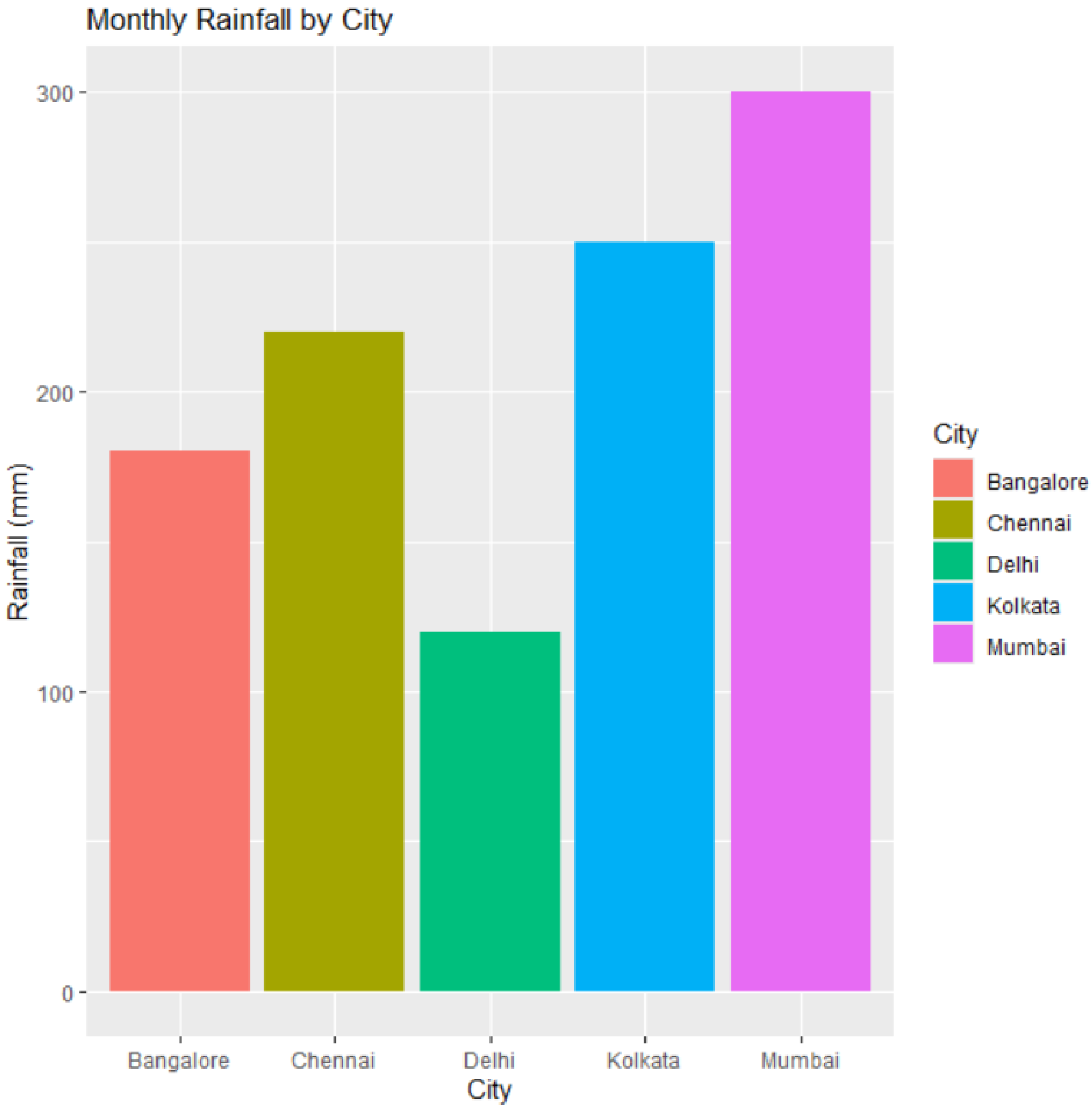
2. Density Plot - Rainfall Distribution



3. Box Plot - Rainfall by City



4. Bar Chart - Rainfall by City



5.Bank Loan Risk Assessment

```r
library(ggplot2)
```

 # Bank Loan Dataset

```r
loan <- data.frame(
  Customer = c("C1","C2","C3","C4","C5","C6","C7","C8","C9","C10"),
  LoanAmount =
c(200000,150000,300000,180000,400000,250000,350000,220000,270000,320000),
  Income = c(50000,40000,80000,45000,90000,60000,85000,55000,65000,75000),
  CreditScore = c(750,680,820,700,850,720,780,690,740,810),
  Repayment =
c("Good","Average","Good","Poor","Good","Average","Good","Poor","Average","Good"))
```

 # 1. Histogram - Loan Amount Distribution

```r
ggplot(loan, aes(x = LoanAmount)) +
  geom_histogram(binwidth = 50000, fill = "skyblue", color = "black") +
  labs(title = "Loan Amount Distribution",
       x = "Loan Amount",
       y = "Frequency")
```

 # 2. Box Plot - Loan Amount by Repayment History

```r
ggplot(loan, aes(x = Repayment, y = LoanAmount, fill = Repayment)) +
  geom_boxplot() +
  labs(title = "Loan Amount by Repayment History",
       x = "Repayment History",
       y = "Loan Amount")
```

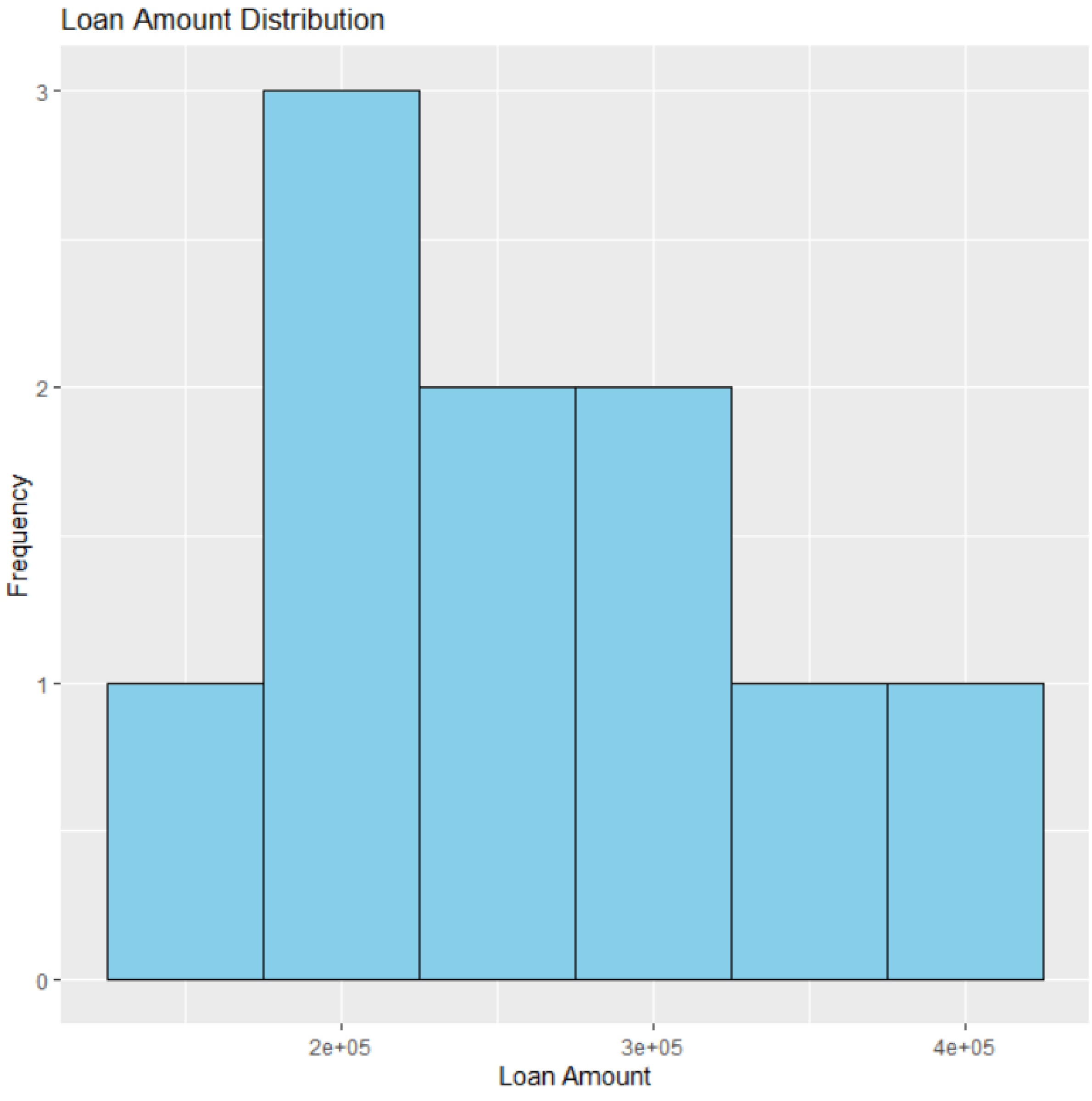
 # 3. Scatter Plot - Income vs Credit Score

```r
ggplot(loan, aes(x = Income, y = CreditScore, color = Repayment)) +
  geom_point(size = 3) +
  labs(title = "Income vs Credit Score",
       x = "Income",
       y = "Credit Score")
```

4. Bar Chart - Customers by Repayment History

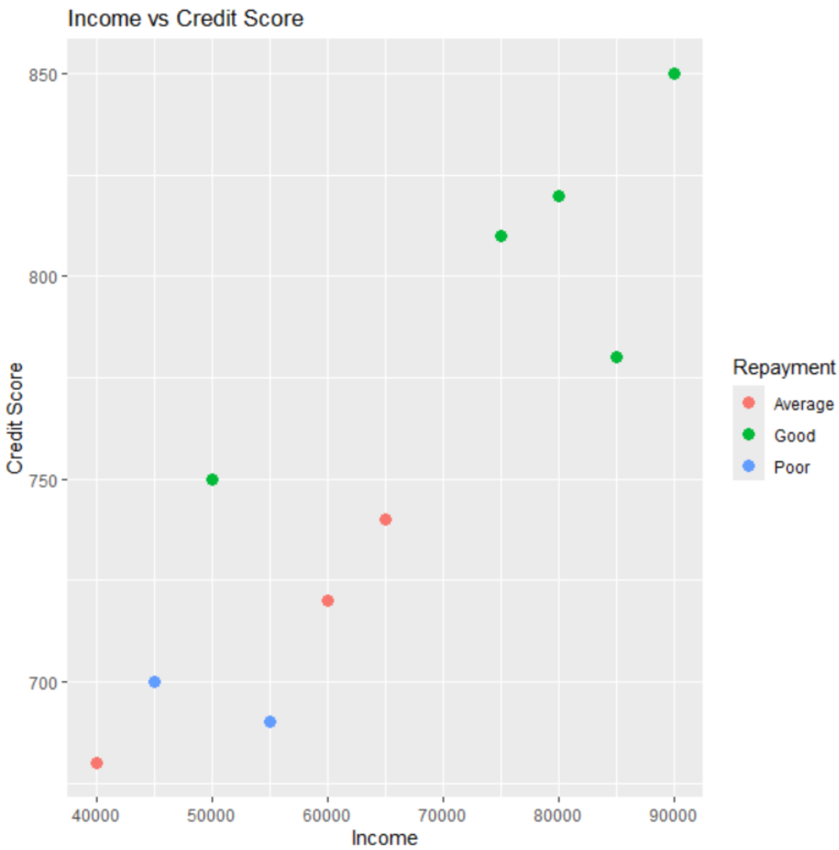
```
ggplot(loan, aes(x = Repayment, fill = Repayment)) +
  geom_bar() +
  labs(title = "Customer Segments by Repayment History",
       x = "Repayment History",
       y = "Number of Customers")
```

1. Histogram - Loan Amount Distribution



2. Box Plot - Loan Amount by Repayment History

3. Scatter Plot - Income vs Credit Score



4. Bar Chart - Customers by Repayment History